1. Business Understanding

Existing Academic Management Process

A typical higher-education institution manages the following processes:
Admission Student Registration Course Registration Attendance
Examination Marks Entry Result Processing Academic Reports
Currently, some institutions may use a combination of:

- Paper records
- Spreadsheets
- Separate departmental systems
- Manual registers
- Email/physical communication

This can create duplication, delays, errors, and difficulty in retrieving information.
Business Opportunity

A centralized academic management platform can:

- Reduce administrative workload.
- Improve data accuracy.
- Provide faster access to student information.
- Automate academic calculations and reports.
- Improve transparency.
- Provide role-based access to sensitive information.
- Create a foundation for future digital academic services.

2. Design Thinking Report

2.1 Empathize

The first stage focuses on understanding users and their difficulties.
Users to understand

- Students
- Faculty members
- Academic administrators
- Examination cell staff
- IT support staff

Research Methods

- Interviews

- Questionnaires
- Observation
- Existing-process analysis
- Feedback from students and faculty

Key Observations

Students need easy access to their academic information.
Faculty members need efficient mechanisms for recording attendance and marks.
Administrators need accurate and consolidated information for decision-making and reporting.

2.2 Define

Problem Statement

Higher-education institutions need a centralized and secure system to manage student academic records because manual and fragmented processes cause data duplication, errors, delays, and difficulty in generating accurate academic information.
User-Centered Problem Statements

Student:

"I need a simple way to view my academic records, attendance, courses, and results without depending on administrative staff."
Faculty:

"I need an efficient system for recording attendance and marks so that I can reduce paperwork and avoid calculation errors."
Administrator:

"I need centralized and accurate student information so that I can efficiently manage academic operations and generate reports."

3. Ideate

Possible solutions include:

1. Centralized student database
2. Role-based login
3. Online student registration
4. Digital attendance management
5. Online marks entry
6. Automatic result calculation

7. Academic report generation
8. Search and filtering
9. Dashboard for different users
10. Audit logs for important operations

Selected Solution

A web-based Student Record and Academic Management Platform with role-based access and centralized academic records.

4. Prototype

The initial prototype should contain:

- Login screen
- Student dashboard
- Faculty dashboard
- Administrator dashboard
- Student registration form
- Student profile
- Course management
- Attendance screen
- Marks-entry screen
- Results screen
- Academic report screen

The prototype can initially be created using wireframes before development begins.

5. Test

Prototype testing should involve representative users.

Testing Questions

- Is the registration process easy to understand?
- Can faculty record attendance efficiently?
- Can students easily find their results?
- Can administrators generate reports?
- Are the navigation and labels clear?
- Are users able to complete tasks without assistance?

Feedback collected during testing will be used to improve subsequent versions.

6. IBM Design Thinking

6.1 Focus on User Outcomes

The system should focus on measurable user outcomes rather than merely implementing features.
Examples:

- Students can quickly access academic records.
- Faculty can record attendance with fewer manual steps.
- Administrators can generate reports faster.
- Examination staff can reduce result-processing errors.

6.2 Restless Reinvention

The product should continuously improve based on:

- User feedback
- Usage data
- New academic requirements
- Technology improvements
- Institutional policies

6.3 Diverse Empowered Teams

The project team should include different perspectives:

- Student representatives
- Faculty representatives
- Administrative staff
- Examination staff
- Developers
- UI/UX designers
- Testing/QA members

This ensures that product decisions are not based only on the developer's perspective.

7. IBM Observe Reflect Make Cycle

Observe

Understand users and their environment.
Example observations:

- Faculty spend considerable time maintaining attendance records.

- Students frequently need information about attendance and results.
- Administrators need consolidated student information.

Reflect

Analyze observations and identify opportunities.
Reflection:

- Centralized records can reduce duplication.
- Automated calculations can reduce errors.
- Dashboards can improve information accessibility.

Make

Create and test solutions.
Proposed solutions:

- Centralized database
- Attendance module
- Result-processing module
- Student dashboard
- Administrator reporting dashboard

The cycle repeats continuously:
Observe Reflect Make Observe Reflect Make

8. Stakeholder Analysis Report

Stakeholder	Role	Needs	Influence
Students	System users	View/manage academic information	High
Faculty	Academic users	Attendance and marks management	High
Academic Administrators	Management	Student and course management	Very High
Examination Cell	Result management	Marks and result processing	Very High
IT Support	Technical support	System maintenance and security	High
Institution Management	Decision makers	Reports and analytics	High

Stakeholder Priority

High Priority

- Students
- Faculty
- Academic administrators
- Examination staff

Supporting Stakeholders

- IT support
- Institution management

9. User Persona Report

Persona 1 – Student

Name: Priya Sharma

Age: 20

Role: Undergraduate Student

Goals

- View personal information
- Check registered courses
- Track attendance
- View examination results
- Download academic reports

Challenges

- Dependence on administrative staff
- Difficulty accessing updated information
- Uncertainty about attendance status
- Delayed result information

Expectations

- Simple interface
- Mobile-friendly access
- Accurate information
- Fast response

Persona 2 – Faculty Member

Name: Rahul Patil

Age: 35

Role: Faculty Member
Goals

- View assigned courses
- Record attendance
- Enter marks
- View student performance

Challenges

- Manual attendance registers
- Repetitive data entry
- Calculation errors
- Time-consuming report preparation

Expectations

- Fast data entry
- Easy navigation
- Automatic calculations
- Reliable system

Persona 3 – Academic Administrator

Name: Anjali Deshmukh

Age: 42

Role: Academic Administrator
Goals

- Manage student records
- Manage courses
- Monitor academic information
- Generate reports

Challenges

- Large volume of student data
- Duplicate records
- Manual report generation
- Maintaining data accuracy

Expectations

- Centralized database
- Powerful search
- Accurate reports
- Role-based security

10. Functional Requirements Specification (FRS)

FR-01: User Authentication

The system shall allow authorized users to log in using appropriate credentials.

FR-02: Role-Based Access

The system shall provide different permissions for:

- Student
- Faculty
- Administrator
- Examination staff

FR-03: Student Registration

The administrator shall be able to:

- Create student records
- Update student information
- View student information
- Search students
- Deactivate student records

FR-04: Student Profile

The system shall maintain:

- Student ID
- Name

- Date of birth
- Contact information
- Program
- Department
- Semester
- Admission information

FR-05: Course Management

Administrators shall be able to:

- Create courses
- Update courses
- Assign courses
- Manage credits
- Associate faculty with courses

FR-06: Attendance Management

Faculty shall be able to:

- View enrolled students
- Mark attendance
- Update attendance where authorized
- View attendance percentage

FR-07: Examination Management

Authorized staff shall be able to:

- Create examinations
- Enter marks
- Update marks
- Verify marks

FR-08: Result Management

The system shall:

- Calculate grades
- Calculate academic performance
- Generate results
- Store historical results

FR-09: Report Generation

The system shall generate:

- Student academic reports
- Attendance reports
- Course reports
- Result reports
- Administrative reports

FR-10: Search

Users with appropriate permissions shall be able to search student and academic records.

FR-11: Audit Trail

The system should record important operations such as:

- Record creation
- Record modification
- Result modification
- User activity

11. Non-Functional Requirements Specification

11.1 Security

- Passwords must be securely stored.
- Access must be role-based.
- Sensitive academic information must be protected.
- User sessions must be managed securely.
- Important actions should be logged.

11.2 Usability

- Simple navigation
- Consistent interface
- Clear error messages
- Responsive design
- Minimal training requirements

11.3 Reliability

The system should:

- Preserve data integrity.
- Minimize service interruptions.
- Handle failures gracefully.
- Support database backup and recovery.

11.4 Performance

The system should provide acceptable response times for common operations such as:

- Login
- Student search
- Attendance entry
- Result viewing
- Report generation

11.5 Scalability

The system should support increasing numbers of:

- Students
- Courses
- Faculty
- Academic departments
- Concurrent users

11.6 Maintainability

The software should use:

- Modular architecture
- Clear coding standards
- Documentation
- Version control
- Automated testing where possible

12. Product Vision Document

Vision Statement

To provide a secure, centralized, user-friendly academic management platform that enables higher-education institutions to efficiently manage student records and deliver accurate academic information to students, faculty, and administrators.

Mission Statement

To digitize and simplify academic record management through reliable software that improves efficiency, accuracy, transparency, and accessibility.

Business Objectives

1. Reduce manual record management.
2. Improve academic data accuracy.
3. Reduce administrative workload.
4. Improve accessibility of student information.
5. Automate academic reporting.
6. Improve data security.
7. Provide a scalable foundation for future academic services.

Success Criteria

The product will be considered successful if it:

- Reduces manual academic record handling.
- Provides centralized student information.
- Allows faculty to efficiently manage attendance.
- Provides accurate academic results.
- Generates reports efficiently.
- Receives positive user feedback.
- Maintains appropriate security and reliability.

13. MVP Definition

MVP Objective

The MVP should provide the minimum set of capabilities required to digitally manage essential student academic records.

MVP Features

Feature	Priority
User Login	Must Have
Role-Based Access	Must Have
Student Registration	Must Have
Student Profile	Must Have
Course Records	Must Have
Attendance Management	Must Have

Marks/Result Management	Must Have
Academic Reports	Must Have
Search	Should Have
Audit Logs	Should Have
Advanced Analytics	Future

MVP User Flow

Login Dashboard Select Module Perform Operation Save View Updated Information

14. Product Roadmap

Phase 1 – Discovery and Planning

Duration: Unit 1
Deliverables:

- Business requirements
- Personas
- Stakeholder analysis
- Design Thinking
- Product vision
- MVP definition
- Initial architecture
- Project charter

Phase 2 – Foundation

Duration: Unit 2
Focus:

- Repository
- Database
- Authentication
- User roles
- Basic UI

Phase 3 – Academic Management

Duration: Unit 3
Focus:

- Student management
- Course management
- Faculty management
- Registration

Phase 4 – Attendance and Examination

Duration: Unit 4
Focus:

- Attendance
- Examination
- Marks
- Result processing

Phase 5 – Reports and Quality

Duration: Unit 5
Focus:

- Academic reports
- Testing
- Security
- Performance
- Bug fixing

Phase 6 – Deployment and Enhancement

Future
Potential features:

- Mobile application
- Notifications
- Analytics dashboard
- Parent portal
- Integration with existing institutional systems
- Online fee management
- AI-assisted academic analytics

15. Initial System Architecture

A suitable initial architecture is a three-tier architecture.

Users

Students | Faculty
Admin | Exam Staff

Presentation Layer

Web UI / Responsive UI

Application Layer

Authentication
Student Management
Course Management
Attendance Management
Examination Management
Result Management
Report Generation

Data Layer

Relational Database
Student Records
Course Records
Attendance
Marks & Results
User Accounts

16. Project Charter

Project Name

Enterprise Student Record and Academic Management Platform
Project Purpose

To develop a centralized platform for managing student records and academic activities.
Project Scope

In Scope

- Student records
- Course records
- Attendance
- Examination/results
- Academic reports
- Authentication
- Role-based access

Out of Scope for MVP

- Fee management
- Hostel management
- Transportation management
- Library management
- Payroll
- Advanced AI analytics

Major Deliverables

- BRD
- FRS
- NFRS
- Product Vision
- Personas
- Stakeholder Analysis
- Design Thinking Report
- MVP
- Roadmap
- Architecture
- Working software in later units

Constraints

- Limited development time
- Limited resources

- Need for data security
- Changing institutional requirements

Risks

Risk	Mitigation
Requirement changes	Maintain prioritized backlog
Data loss	Regular backups
Security issues	Authentication and authorization
Poor usability	User testing
Scope creep	Clearly define MVP
Development delays	Agile milestones